

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0708

Tinkertown 138 kV
January 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0708. The Interconnection Customer proposes a 31.6 MW solar photovoltaic facility interconnecting at the Tinkertown 138 kV substation in the Kentucky area. The request was received on November 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 222 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,958,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dewey / Tillamook Substation 138 KV Line	Baker / Hubbardstown 345 KV	141.8	191	19.1
Pocket / Tillamook 138 KV L	Hubbardstown / Baker 345 KV	137.6	191	19.1
Dewey / Dowell 138 KV	Baker / Hubbardstown 345 KV	132.4	210	19.1
Mt Vernon / Pocket 138 KV	Baker / Hubbardstown 345 KV	124.7	203	19.1
Hazard / Harris 138 KV line	Wansley Combined Cycle / Massapequa 138 KV	114.5	227	7.1
Sugar Hill / AlleghenyEnergy Units 3 4 & 5 345 KV Line	Piney Crk Project / Hilltop 500 KV	112.8	1,438	6.2
Allegheny Energy Units 3 4 & 5 345 kV/1 kV XFMR	Hilltop / Piney Crk Project 500 KV	111.8	1,302	6.2
Allegheny Energy Units 3 4 & 5 500 kV/1 kV Auto	Piney Creek Project / Hilltop 500 KV Line	111.8	1,302	6.2
Spring Crk Rd / Harris 138 KV	Wansley Combined Cycle / Massapequa 138 KV Line	110.3	315	14.7
Tinkertown / Stillwater B 138 KV L	Davisville / North Appleton 138 KV	109.4	208	22.3
Meeker / Sidney Coal 138 KV	Spring Creek Rd / Harris 138 KV line	103.8	231	6.3
Meekr / Turkey Crk 138 KV	Spring Crk Rd / Harris 138 KV	103.3	219	6.3
Sporn / Beck Road 230 KV	Beck Road / Sporn 345 KV	102.6	508	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tillamook / Dewey 138 KV	Rebuild line with higher-capacity conductor	\$11,899,000	7% (\$859,000)
Pocket / Tillamook 138 KV L	Rebuild line with higher-capacity conductor	\$6,474,000	7% (\$467,000)
Dewey / Dowell 138 KV	Rebuild line with higher-capacity conductor	\$6,195,000	7% (\$447,000)
Mt Vernon / Pocket 138 KV	Rebuild line with higher-capacity conductor	\$4,698,000	7% (\$339,000)
Hazard / Harris 138 KV line	Rebuild line with higher-capacity conductor	\$34,513,000	1% (\$308,000)
Sugar Hill / AlleghenyEnergy Units 3 4 & 5 345 KV Line	Rebuild line with higher-capacity conductor	\$22,179,000	0% (\$82,000)
Allegheny Energy Units 3 4 & 5 345 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$17,812,000	0% (\$66,000)
Allegheny Energy Units 3 4 & 5 500 kV/1 kV Auto	Replace transformer with higher-rated unit	\$35,356,000	0% (\$132,000)
Spring Creek Road / Harris 138 KV	Rebuild line with higher-capacity conductor	\$2,888,000	1% (\$28,000)
Tinkertown / Stillwater B 138 KV L	Reconductor line	\$2,659,000	7% (\$177,000)
Meeker / Sidney Coal 138 KV	Reconductor line	\$3,069,000	1% (\$30,000)
Meeker / Turkey Crk 138 KV	Reconductor line	\$193,000	1% (\$2,000)
Sporn / Beck Road 230 KV	Reconductor line	\$2,734,000	1% (\$21,000)
	Total allocated to project		\$2,958,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.